**Worksheet 6.1**

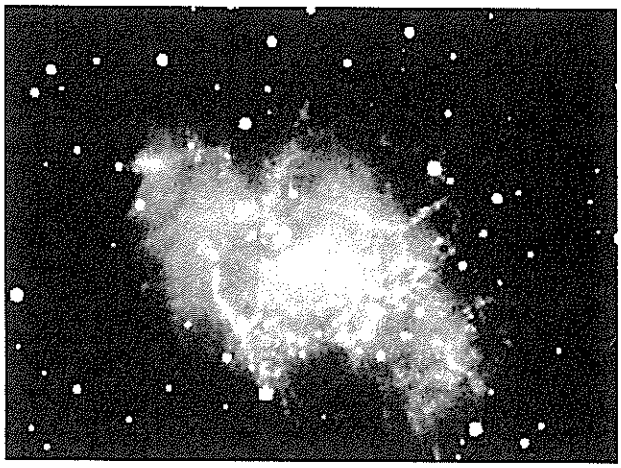
*Science Quest 10:* pages 232–3

# Observing stars

Student: ..... Class: .....

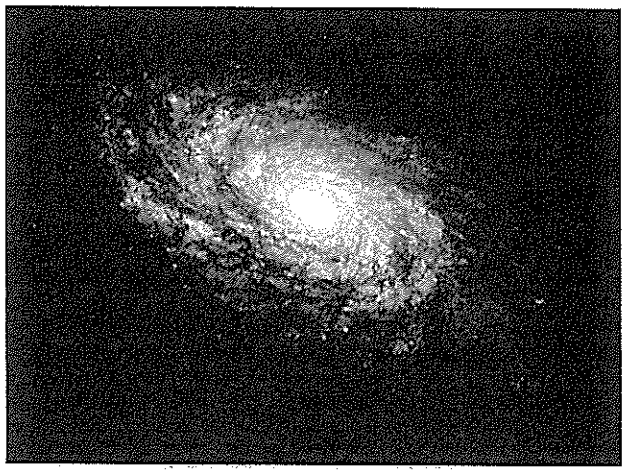
**1.** Three types of celestial objects are illustrated below. Identify and outline the nature of each.

(a)



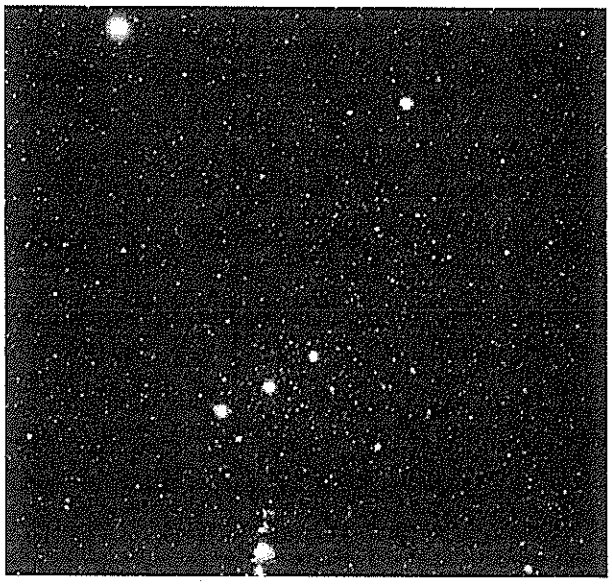
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(b)



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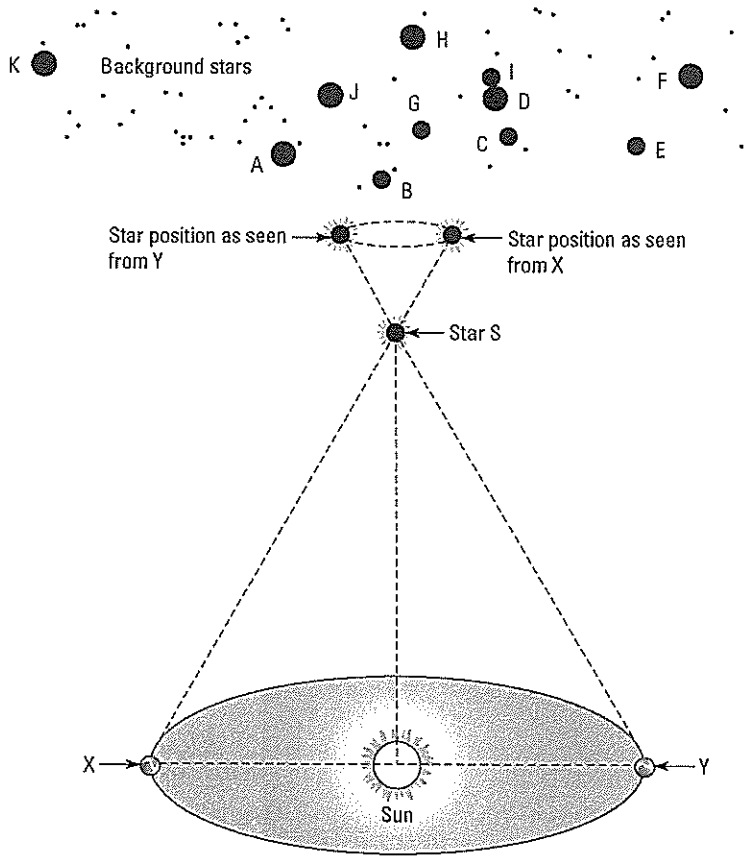
(c)



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**2.** The following diagram shows the effect of parallax on the observation of a close star (S) from two locations (X and Y) in the Earth's orbit around the Sun.




(a) The Earth is shown in two positions X and Y. How long does it take for the Earth to move from X to Y?

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(b) Explain the meaning of parallax in relation to the apparent motion of stars.

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(c) Before 1837 it was believed that all stars existed on a transparent celestial sphere at a fixed distance from the Earth. How did the observation of parallax of close stars change this view?

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(d) Outline the motion of the stars as seen by an observer on Earth during the time taken to travel from X to Y.

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